

Step 2 of the scope-completion roadmap: the additive-energy extremal (EXT)

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

Step 2 addresses the competitor-class widening from the lattice Reading-H class to the full admissible class $\mathcal{C}_{\text{full}}$ (lattice + non-lattice). The off-diagonal stability is the ratio

$$R_{\text{lead}}(Q) = 23.2 (1 + K_{\text{floor}}(Q)) I, \quad R_{\text{lead}} < 1 \iff K_{\text{floor}} < K_* = 20.55,$$

and the analysis turns on *not* conflating two quantities:

$$T'(Q) = \max_{t \neq 0} \#(Q \cap C_t) \quad (L^\infty \text{ occupancy}), \quad K_{\text{floor}}(Q) = \frac{1}{N^2} \sum_{t \neq 0} r(t)^2 \quad (L^2 \text{ additive-energy floor}),$$

with $K_{\text{floor}} \leq T'$ unconditionally (R-025: $E_+ \leq (1 + T')N^2$).

2. The elementary antipodal lemma (proved)

Lemma. For any finite antipodal set $Q = -Q$ (N points) and any $t \neq 0$, $\#(Q \cap C_t) \leq N/2$, where $C_t = \{x : x \cdot t = |t|^2/2\}$.

Proof. If $x \in C_t$ then $(-x) \cdot t = -|t|^2/2 \neq |t|^2/2$ (as $t \neq 0$), so $-x \notin C_t$. Each antipodal pair $\{x, -x\}$ therefore contributes at most one point to C_t ; with $N/2$ pairs, $\#(Q \cap C_t) \leq N/2$. ■

Hence $T'(Q) \leq N/2$ for every antipodal competitor, and $K_{\text{floor}} \leq T' \leq N/2$.

3. The off-diagonal $\mathcal{C}_{\text{full}}$ closure is PROVED (thin)

The admissible class is coherence-resolved, so $N \leq n_{\text{pack}} = 16/\theta^2 = 40.68$ at the operating point. Therefore, for EVERY admissible competitor (lattice OR non-lattice),

$$K_{\text{floor}} \leq T' \leq \frac{N}{2} \leq \frac{n_{\text{pack}}}{2} = 20 < 20.55 = K_*, \quad \Rightarrow \quad R_{\text{lead}} \leq 23.2 (1 + 20) (2 \times 10^{-3}) = 0.974 < 1.$$

This is the A_ext universal fallback of T-022/T-023: **the off-diagonal sector for $\mathcal{C}_{\text{full}}$ is PROVED**, with a THIN margin $\times 1.026$. It is NOT a conjecture. The lattice restriction of the enacted T7 ($T' \leq 13 \Rightarrow$ margin $\times 1.5$) bought the COMFORTABLE margin, not the closure.

4. The EXT conjecture: a margin upgrade, not a blocker

The probe res5_033 maps the ACTUAL floor across the admissible families:

family (non-lattice unless noted)	N	T'	K_{floor}
random antipodal	20–40	2	0.85–0.93
latitude-ring antipodal pairs (T' maximal)	20–40	$N/2$ (10–20)	2.05–2.27
rich $\mathbb{Z}^3$ lattice shells	30–36	8	2.34–2.47

The L^∞ occupancy T' reaches its maximum $N/2 = 20$ (latitude rings) while the L^2 floor K_{floor} stays ≤ 2.5 : **T' is a loose proxy for K_{floor}** . The EXT conjecture — $\max_{\mathcal{C}_{\text{full}}} K_{\text{floor}}$ is attained in the arithmetic subclass at ~ 3 — would replace the loose bound $K_{\text{floor}} \leq N/2 = 20$ by the tight $K_{\text{floor}} \leq 3$, upgrading the off-diagonal margin from the thin $\times 1.026$ to a comfortable

$$R_{\text{lead}} \leq 23.2 (1 + 3) (2 \times 10^{-3}) = 0.19 \quad (\text{margin} \times 6).$$

So EXT is a *margin-comfort upgrade*, NOT a closure blocker. Per the operator decision (2026-06-10) it remains **T2 CONJECTURE** (strong evidence; falsifier: an admissible Q with $K_{\text{floor}} > 13$); promotion to T6/T7 requires a tight L^2 additive-energy bound (arithmetic-maximises-energy, a Freiman-type extremal statement).

5. Self-audit correction to res5_031 (CLAUDE.md 6.3)

res5_031's headline “the lattice is the off-diagonal worst case” used the SMALL BCC 12-shell ($K_{\text{floor}} = 1.75$ excl). The broader map here shows rich lattice shells (~ 2.5) AND latitude-ring antipodal pairs (~ 2.3) both EXCEED the 12-shell. The honest statement is: $K_{\text{floor}} \sim 1\text{--}3$ across ALL admissible classes, far below K_* , with T' a loose proxy — NOT “lattice strictly worst”. res5_031's per-family asserts remain valid; only the comparative headline is corrected.

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “*The thin closure rests on $n_{\text{pack}} \leq 41$.*” VALID and stated: $n_{\text{pack}} = 40.68$ at the operating point and DECREASES with μ^2 (larger $\hat{r}$), so $N/2 \leq 20 < 20.55$ holds across the certified window; the $\times 1.026$ margin is thin but rigorous. EXT removes the thinness.
- (β) “*Does this widen the enacted T7?*” No — it proves the off-diagonal piece for $\mathcal{C}_{\text{full}}$ (thin), but the full $\mathcal{C}_{\text{full}}$ T7 also needs the (D) and (S) sectors for non-lattice (Step 3) and an operator decision on accepting the thin off-diagonal margin. No B1/B2 tier change.
- (γ) “*Is the antipodal lemma circular with R-025?*” No — the lemma bounds T' (occupancy) by elementary geometry; R-025 bounds K_{floor} by T' (additive combinatorics). Independent inputs composed.

Quantitative sanity check (mandatory): *lemma $T' \leq N/2$* verified for all families; *thin closure $K_{\text{floor}} \leq 20 < 20.55$* , $R_{\text{lead}} \leq 0.974 < 1$; *actual $K_{\text{floor}} \leq 2.47$* , $R_{\text{lead}} \leq 0.16$ (margin $\times 6$). All from res5_033 (5/5).

7. Result footer

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Result ID:          B1/B2 EXT status + off-diagonal C_full thin
                    closure (roadmap Step 2)
Precise statement:  Antipodal lemma  $T' \leq N/2$  (proved, elementary).
                    With R-025 ( $K_{\text{floor}} \leq T'$ )
                    and packing  $N \leq n_{\text{pack}} = 40.68$ :
                                 $K_{\text{floor}} \leq N/2 \leq 20 < 20.55 = K_*$ 
                                 $\Rightarrow R_{\text{lead}} < 1$  for
                    ALL admissible competitors incl non-lattice
                    (THIN  $\times 1.026$ , A_ext
                    fallback) -- the off-diagonal C_full closure is
                    PROVED, not conjecture.
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Actual $K_{\text{floor}} \sim 1-2.5$ across
random/latitude-ring/rich-shell (T' up to
 $N/2=20$ a loose proxy). EXT (max $K_{\text{floor}}=3$ in
arithmetic subclass) is a
MARGIN UPGRADE $x1.026 \rightarrow \sim x6$, NOT a closure
blocker; stays T2.

Scope: Off-diagonal sector for C_{full} . (D)/(S)
non-lattice = Step 3; thin-margin
acceptance = operator decision. No B1/B2 tier
change.

Dependencies: antipodal lemma (this note, proved); R-025
($K_{\text{floor}} \leq T'$, T7); T-018
($R_{\text{lead}}/K \cdot \text{identity}$); T-022/T-023 (A_{ext}
fallback); res5_031 (corrected).

Evidence grade: off-diagonal C_{full} thin closure PROVED (T6,
on R-025+packing);
EXT T2 CONJECTURE (strong evidence). EXECUTED
(res5_033 5/5).

Reproduction command: python
codes/vacuum/res5_033_ext_adversarial_map.py

Expected output: antipodal lemma $T' \leq N/2$ verified;
 $R_{\text{lead}} \leq 0.974 < 1$ (C_{full} thin);
actual $K_{\text{floor}} \leq 2.47$ (margin $x6$); 5/5 PASS.

Falsifier: EXT: an admissible Q with $K_{\text{floor}} > 13$. Thin
closure: an admissible
antipodal Q with $T' > N/2$ (would refute the
lemma) or $N > n_{\text{pack}}$.

Tier before / after: No B1/B2 change. Antipodal lemma proved;
off-diagonal C_{full} thin
closure proved (T6); EXT T2 (margin upgrade, operator:
do
not
promote).

No-overclaim statement: The off-diagonal C_{full} closure is THIN
($x1.026$) and rests on
 $n_{\text{pack}} \leq 41$; the full C_{full} T7 also needs
(D)/(S) non-lattice (Step 3).
EXT is a comfort upgrade, not the closure. No
tier widening.

Next required action: Step 3 -- extend (D) diagonal isotropy + (S)
selection floor to
non-lattice competitors; then operator decision
on thin-margin C_{full} .